

A Project Report

On

EMBERNET

Offline-First Disaster Alert System Using Bluetooth Mesh Networking

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1. Executive Summary

Natural disasters, floods, and large-scale emergencies consistently expose a critical flaw in modern emergency response systems: they depend entirely on functioning telecommunications infrastructure. When cell towers go down, Wi-Fi routers fail, and the power grid collapses, conventional smartphone applications become useless precisely when they are needed most.

EmberNet is an Android application designed to solve this paradigm architectural bottleneck by transforming every nearby smartphone into an autonomous node within a peer-to-peer (P2P) mesh network. Utilizing Bluetooth Low Energy (BLE) advertising and Wi-Fi Direct protocols, the application broadcasts critical SOS signals containing the sender's precise GPS coordinates and real-time battery level. Each receiving device automatically relays the message to the next, propagating data dynamically across the mesh until the alert successfully reaches an active gateway or responder capable of summoning assistance.

Furthermore, EmberNet embeds a complete, localized offline mapping system directly on the device filesystem. Pre-downloaded while an internet connection remains active, these mapping layers utilize OpenStreetMap data via the OSMDroid framework. This allows victims and disaster rescue personnel to visualize critical coordinates, civilian safe zones, local hospitals, and designated hazard zones without requiring cellular or internet connectivity.

2. Problem Statement & Market Analysis

2.1 The Core Infrastructure Problem

Every major natural disaster over the past decade has exposed a structural vulnerability: populations requiring urgent humanitarian assistance lose access to communications networks immediately. Telecommunication towers suffer from load saturation, structural damage, or power grid failures, rendering mainstream computational devices useless when lives are on the line. In these scenarios, expensive mobile smartphones are functionally reduced to local flashlights.

2.2 Target Demographic and Vulnerable Groups

- **Disaster Victims:** Individuals trapped in collapsed architectural infrastructure or isolated by rising floodwaters who are unable to reach physical network coverage.
- **Remote/Rural Populations:** Communities with structurally fragile baseline telecom frameworks that are highly susceptible to early disconnects during catastrophic events.
- **Off-Grid Travelers:** Hikers, mountaineers, and field researchers moving outside the operational boundaries of typical civilian telecommunication grids.

2.3 Analytical Comparison of Existing Frameworks

Emergency Solution	Architectural Limitation	The EmberNet Advantage
SMS / Telephony Calls	Requires localized cell tower functionality and structural network availability.	Operates effectively with zero active cellular network signals.
WhatsApp / Signal	Dependent on TCP/IP connections and central internet gateway access.	Utilizes a fully offline, self-healing peer-to-peer mesh architecture.
Satellite Telephony	Prohibitively expensive, specialized, and rarely deployed among general populations.	Deploys directly on standard consumer smartphones via built-in radios.
Google Maps Offline	Provides read-only spatial data with no peer-to-peer transmission capabilities.	Integrates offline maps alongside real-time peer-to-peer SOS alert routing.

3. System Objectives

3.1 Primary Objectives (Must-Have Requirements)

- BLE Mesh Implementation: Engineer a decentralized Bluetooth Low Energy mesh routing network allowing emergency SOS payloads to hop over devices seamlessly without internet dependencies.
- SOS Broadcast Engine: Design a background serialization module that binds real-time GPS geographic coordinates and precise battery telemetry to each outbound alert packet.
- High-Priority UI Overlay: Program an assertive incoming alert handler featuring a full-screen window overlay, distinctive device vibration sequences, and continuous system notifications.
- Airplane Mode Enforcement: Ensure the entire end-to-end operation remains stable under strict Airplane Mode configurations with localized Bluetooth and Wi-Fi radios enabled manually.

3.2 Secondary Objectives (Nice-to-Have Enhancements)

- Wi-Fi Direct Subsystem: Integrate Wi-Fi P2P capabilities to extend payload transport distance beyond the physical limits ($d \leq 50m$) inherent to baseline BLE configurations.
- Persistent Alert Logging: Establish a local data repository to maintain a clean history log of incoming and outgoing alerts across application reboots.

- Spatial Safe-Zone Anchors: Implement dynamic visual markers on the offline map tracking active regional hospitals, medical units, and temporary emergency shelters.
- Foreground Execution Engine: Implement an Android ForegroundService model to guarantee radio broadcasting activities are not throttled when the smartphone screen is inactive.

4. Scope of Work & System Boundaries

4.1 Features Inside System Scope

- Wi-Fi Direct Protocols: Peer discovery mechanisms and raw bytes payload transfers over Wi-Fi P2P groups.
- Visual Polish: Modern fluid user interface containing an interactive Lottie-animated splash screen and intuitive SOS broadcast animations.
- Telemetry Visualization: Dynamic color-coded battery status tracking (Green → Amber → Red) embedded inside the central SOS interface.
- Asset Downloading Dialogs: Custom 'Ember Progress Bar' component embedded within an active map tile pre-caching progress modal.
- Live GPS Telemetry: Real-time hardware tracking using hardware location providers, projecting a persistent blue location dot on the map.
- Hop-Chain Visualization: Visual diagnostic screen rendering the path an incoming alert followed across the node mesh.
- Local Cache Persistence: Safe storage of chronological alert histories using key-value serialization via SharedPreferences.

5. Technical Specifications & Stack Architecture

5.1 Implementation Constraints & Radio Boundaries

The underlying mesh topology behaves according to strict physical and operating system limitations. The maximum operational range of a single BLE transmission hop is mathematically constrained to approximately $R \approx 50$ meters in open, unobstructed space. Furthermore, the standard BLE advertising payload size is restricted to a raw limit of 17 usable bytes.

To bypass this limitation, EmberNet utilizes advertising packets strictly for peer discovery and alert signaling. Once an alert state is recognized, the system automatically initializes a robust BluetoothGatt connection to transfer the complete, unrestricted metadata payload.

From an operating system perspective, EmberNet requires a minimum of Android API level 26 (Android 8.0) to correctly process modern runtime split-permission frameworks for fine-grained location tracking and Bluetooth scanning. Additionally, testing can only be executed via physical

test devices, as standard emulator hypervisors lack the low-level hardware virtualization layers required to accurately simulate Bluetooth mesh topologies and Wi-Fi Direct negotiations.

5.2 Chosen Technology Stack

Category	Selected Technology	Operational Purpose within Architecture
IDE	Android Studio	Primary Development Environment for compilation and resource bundling.
Backend Language	Java (JDK 17)	Core algorithmic framework, thread management, and radio logic.
User Interface Kit	Material Design 3 (MDC)	Ensures responsive UI utilizing high-contrast theme layout elements.
Animation Engine	Lottie Android (6.4.0)	Renders high-fidelity splash frames and reactive waveforms.
Mapping Framework	OSMDroid (6.1.18)	Handles local spatial queries and memory-efficient offline map tiles.
Map Raster Assets	OpenStreetMap Free Tiles	Allows unthrottled, zero-cost caching of local maps without API keys.
Geospatial Provider	FusedLocationProvider	Extracts accurate coordinate telemetry directly via hardware GPS.
Data Link Layer	BluetoothGattServer / Client	Manages state connections and byte-stream delivery across discovered peers.

6. Conclusion

EmberNet demonstrates that life-saving emergency communication does not require expensive satellite hardware, licensed radio equipment, or functioning cellular infrastructure. By combining technologies already present in every modern Android smartphone—Bluetooth Low Energy, Wi-Fi Direct, GPS, and local storage—EmberNet creates a resilient mesh network that activates precisely when conventional systems fail. It is a technical project engineered with deep architectural intent, built not just for a grade, but for when everything goes wrong.